

Cover Letter for JMLR Submission

Date: February 26, 2026

To: JMLR Editorial Board

Re: Submission of “Engagement as Entanglement: Variance Signatures of Bidirectional Context Coupling in Large Language Models”

Dear Editors,

I am pleased to submit the manuscript titled “Engagement as Entanglement: Variance Signatures of Bidirectional Context Coupling in Large Language Models” for consideration for publication in the Journal of Machine Learning Research.

1. Summary of Contribution

This paper introduces an entanglement framework that provides mechanistic explanation for the “Lost in Conversation” phenomenon documented by Laban et al. (2025), who showed that LLMs exhibit 112% increased unreliability across 200,000+ multi-turn conversations. We demonstrate that context sensitivity (ΔRCI) tracks variance reduction in response embeddings ($r = 0.76$, $p = 2.37 \times 10^{-68}$, $N = 360$), revealing bidirectional context coupling: convergent entanglement (context stabilizes) and divergent entanglement (context destabilizes). The Entanglement Stability Index (ESI) predicts which models will fail in multi-turn settings.

2. Prior Publications

Earlier versions of foundational concepts appear in the following preprints:

- Laxman (2026a). “Context Curves Behavior: Measuring AI Relational Dynamics with ΔRCI .” Preprints.org
DOI: 10.20944/preprints202601.1881.v2
- Laxman (2026b). “Standardized Context Sensitivity Benchmark.” Preprints.org
DOI: 10.20944/preprints202602.1114.v2

Differences from prior work: The current submission introduces the entanglement framework, variance analysis, bidirectional coupling theory, and the ESI metric—none of which appear in the preprints. The preprints establish the ΔRCI metric; this paper explains its mechanistic basis. This represents substantial (>50%) new contribution.

3. Co-author Consent

I am the sole author of this manuscript.

4. Conflicts of Interest

I have no conflicts of interest with any JMLR action editors or reviewers.

5. Suggested Action Editors

Based on research alignment with the current JMLR Action Editor list, I suggest the following action editors:

1. Kai-Wei Chang (UCLA) — Large Language Models, Trustworthy NLP, Vision-Language Models

2. Qiang Liu (University of Texas at Austin) — Deep learning, large language models, generative models
3. Quanquan Gu (UCLA) — LLMs, deep learning theory, optimization

6. Suggested Reviewers

1. Philippe Laban (Salesforce AI) — author of “Lost in Conversation” paper
2. Nelson Liu (Stanford) — author of “Lost in the Middle” paper
3. Nils Reimers (Hugging Face) — sentence embeddings expert
4. Tatsunori Hashimoto (Stanford) — LLM evaluation
5. Sara Hooker (Cohere) — model evaluation and safety

7. Keywords

Large language models, context sensitivity, multi-turn conversation, variance analysis, AI evaluation, model stability, entanglement

8. Data and Code

All data and code are publicly available at: <https://github.com/LaxmanNandi/MCH-Research>

Thank you for considering this submission.

Sincerely,

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